COMMERCIAL ITEM DESCRIPTION

MIXER, DENTAL CASTING INVESTMENT (Mechanical)

The General Services Administration has authorized the use of this commercial item description in preference to Interim Federal Specification GG-M-00500A (DSA-DM).

Salient characteristics:

This commercial item description covers a mechanical dental mixer suitable for mixing gypsum products and investment materials.

The mixer shall be a hand-operated, mechanical mixer suitable for use in mixing dental gypsum products and investment materials.

Design and construction. The dental casting mixer shall conform to Defense Personnel Support Center (DPSC) Drawing 10088 for style, design, and dimensions, except drilling through shall be optional for component gears. The mixer shall be compact, easy to handle, and free turning when tested as specified herein. The bottom portion of the cover shall fit snugly into the mixing bowl. A flange on the extreme outside edge of the cover shall fit down over the bowl. The paddle shall be shaped to fit the inside contour of the bowl. The paddle shall be permanently attached to the pin by soldering, brazing or welding and shall clear the walls and bottom of the bowl without interference. The gear housing shall be waterproof. The gear housing cover shall be held in place with screws as indicated in the drawing. The gears shall be sealed in a water-tight compartment. The gear ratio of the mixer shall provide 3 to 4 revolutions of the paddle for each turn of the handle.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:22Z
Check the source to verify that this is the current version before use.

Material. The dental casting mixer components shall be fabricated of the materials shown below.

Component	Material	Plating
Paddle Pin	Type 303 corrosion-resisting steel in accordance with ASTM F899 for composition only.	-
Knob-drive Pin	Brass alloy No. 360 in accordance with ASTM B16 for composition only.	Chromium
Paddle	Type 302 or 304 corrosion-resisting steel in accordance with ASTM F899 for composition only.	-
Gear housing and housing cover	Die cast zinc alloy AG40A in accordance with ASTM B86 for composition only.	Chromium
Handle	Brass alloy No. 260 in accordance with ASTM B36 for composition only.	Chromium
Bowl Cover	Die cast zinc alloy AG40A in accordance with ASTM B86 for composition only.	Chromium
Bushings	Type 12L14 carbon steel in accordance with ASTM A108 for composition only.	Zinc
Pinion gear	Naval bronze alloy 464 in accordance with ASTM B21 for composition only.	-
Drive gear	Acetal type thermoplastic.	-
Knob	Brass alloy No. 360 in accordance with ASTM B16 for composition only.	Chromium
Bowl	Vinyl elastomeric material. 1/	-

1/ The specific gravity of the elastomeric material shall be 1.21 to 1.29 when tested as specified in ASTM D792.

Hardness. The hardness (Durometer Share A) of the elastomeric material shall be 54 to 78 when tested as specified in ASTM D2240.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

Finish. All exposed surfaces of bowl cover, handle, gear housing, gear housing cover, knob, and knob drive pin shall be chromium plated. The coatings shall have a bright finish except where textured, and shall be adherent and free from blisters, pits and other surface defects. All other exposed non-stainless steel metal surfaces shall be treated with a suitable corrosion-resistant finish in accordance with the manufacturer's commercial practice.

Identification marking. Each mixer shall be legibly stamped with the name or registered trademark of the manufacturer.

Workmanship. The bowl shall be free from blisters, cracks, voids, embedded foreign matter or other defects which may impair its function. The finished mixer shall be free from flaking, pitting, scratches or any other other defects which may impair its proper function.

Unit. Each (EA). One mixer, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one assembled mixer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

Unless otherwise specified, sampling for examination shall be conducted at inspection level II, except dimension shall be conducted at inspection level S-1, AQL 2.5 (percent defective), and sampling for tests (when applicable) shall be conducted at inspection level S-1. For tests, acceptance number shall be zero (0) for all sample sizes.

Examination. Examination shall include but shall not be limited to the following:

Plating shows signs of peeling and flaking.
Mixer surfaces corroded.
Burs and slivers present on mixer edges.
Gears do not mesh properly.
Brazing cracked and not ground smooth on paddle and pin.
Handle not securely attached to mixer.
Dimensions affecting fit of mixer and bowl not as specified.
Dimensions affecting mesh of gears not as specified.
Gear ratio not as specified.
Identification marking not as specified.
Set screws not recessed properly in gear housing cover.
Cuts and tears present in bowl.
Foreign matter present in bowl.

Tests.

Operation test. The mixer handle shall be given 100 dual revolutions, in an alternate clockwise and counterclockwise direction and at the rate of 50 revolutions per minute to check the mixers freedom from binding and friction in the gear train or drive shaft bushing. No appreciable heat shall be permitted in the gear box or bushing as a result of this test and there shall be no evidence of gear lubricant present on the paddle pin adjacent to its brass bushing. There shall also be no interference permitted in the mixers operation due to any of the paddle components fouling the inner surface of the mixing bowl.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-00-541-9400	MIXER, DENTAL CASTING INVESTMENT (Mechanical)

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:22Z
Check the source to verify that this is the current version before use.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2089

48076

Location: NCR Disk#6

☆U.S. GOVERNMENT PRINTING OFFICE: 1989 - 604-033/10088

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN: DPSC-RSTE
2800 South 20th Street
PHILADELPHIA,PENNSYLVANIA 19101-8419

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53638	MIXER, DENTAL CASTING INVESTMENT (Mechanical)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:22Z
Check the source to verify that this is the current version before use.